

W3 – 4.5 Double Angle Formulas

MHF4U

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1) Express each of the following as a single trig ratio.

a) $2 \sin(5x) \cos(5x)$

$$= \sin[2(5x)]$$
$$= \sin(10x)$$

b) $\cos^2 \theta - \sin^2 \theta$

$$= \cos(2\theta)$$

c) $1 - 2 \sin^2(3x)$

$$= \cos[2(3x)]$$
$$= \cos(6x)$$

d) $\frac{2 \tan(4x)}{1 - \tan^2(4x)}$

$$= \tan[2(4x)]$$
$$= \tan(8x)$$

e) $4 \sin \theta \cos \theta$

$$= 2 \times 2 \sin \theta \cos \theta$$
$$= 2 \sin(2\theta)$$

f) $2 \cos^2 \frac{\theta}{2} - 1$

$$= \cos[2(\frac{\theta}{2})]$$
$$= \cos(\theta)$$

2) Express each of the following as a single trig ratio and then evaluate

a) $2 \sin 45^\circ \cos 45^\circ$

$$= \sin[2(45^\circ)]$$
$$= \sin(90^\circ)$$
$$= 1$$

b) $\cos^2 30^\circ - \sin^2 30^\circ$

$$= \cos[2(30^\circ)]$$
$$= \cos(60^\circ)$$
$$= \frac{1}{2}$$

c) $2 \sin \frac{\pi}{12} \cos \frac{\pi}{12}$

$$= \sin[2(\frac{\pi}{12})]$$
$$= \sin(\frac{\pi}{6})$$
$$= \frac{1}{2}$$

d) $\cos^2 \frac{\pi}{12} - \sin^2 \frac{\pi}{12}$

$$= \cos[2(\frac{\pi}{12})]$$
$$= \cos(\frac{\pi}{6})$$
$$= \frac{\sqrt{3}}{2}$$

e) $1 - 2 \sin^2 \frac{3\pi}{8}$

$$= \cos[2(\frac{3\pi}{8})]$$
$$= \cos(\frac{3\pi}{4})$$
$$= -\cos(\frac{\pi}{4})$$
$$= -\frac{\sqrt{2}}{2}$$

f) $2 \tan 60^\circ \cos^2 60^\circ$

$$= 2 \left(\frac{\sin 60^\circ}{\cos 60^\circ} \right) \cos^2(60^\circ)$$
$$= 2 \sin 60^\circ \cos 60^\circ$$
$$= \sin[2(60^\circ)]$$
$$= \sin(120^\circ)$$
$$= \sin(60^\circ)$$
$$= \frac{\sqrt{3}}{2}$$

3) Use a double angle formula to rewrite each trig ratio

a) $\sin(4\theta)$

$$= \sin[2(2\theta)]$$
$$= 2 \sin(2\theta) \cos(2\theta)$$

b) $\cos(3x)$

$$= \cos[2(\frac{3}{2}x)]$$
$$= \cos^2(\frac{3}{2}x) - \sin^2(\frac{3}{2}x)$$

c) $\tan x$

$$= \tan[2(\frac{1}{2}x)]$$
$$= \frac{2 \tan(\frac{1}{2}x)}{1 - \tan^2(\frac{1}{2}x)}$$

d) $\cos(6\theta)$

$$= \cos[2(3\theta)]$$
$$= \cos^2(3\theta) - \sin^2(3\theta)$$

e) $\sin x$

$$= \sin[2(\frac{1}{2}x)]$$
$$= 2 \sin(\frac{1}{2}x) \cos(\frac{1}{2}x)$$

f) $\tan(5\theta)$

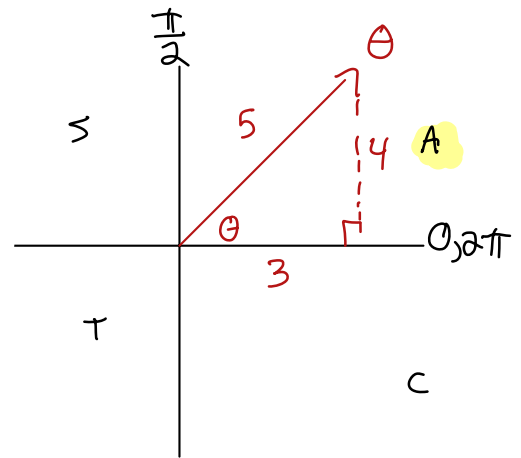
$$= \tan[2(\frac{5}{2}\theta)]$$
$$= \frac{2 \tan(\frac{5}{2}\theta)}{1 - \tan^2(\frac{5}{2}\theta)}$$

4) Determine the values of $\sin 2\theta$, $\cos 2\theta$, and $\tan 2\theta$, given $\cos \theta = \frac{3}{5}$ and $0 \leq \theta \leq \frac{\pi}{2}$

$$\sin(2\theta) = 2 \sin \theta \cos \theta = 2 \left(\frac{4}{5}\right) \left(\frac{3}{5}\right) = \frac{24}{25}$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = \left(\frac{3}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2 \left(\frac{4}{3}\right)}{1 - \left(\frac{4}{3}\right)^2} = \frac{\left(\frac{8}{3}\right)}{\left(-\frac{7}{9}\right)} = \left(\frac{8}{3}\right) \left(-\frac{9}{7}\right) = -\frac{72}{21} = -\frac{24}{7}$$

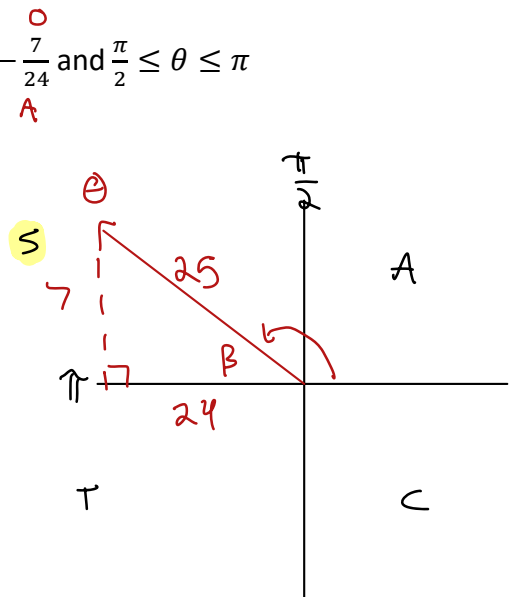


5) Determine the values of $\sin 2\theta$, $\cos 2\theta$, and $\tan 2\theta$, given $\tan \theta = -\frac{7}{24}$ and $\frac{\pi}{2} \leq \theta \leq \pi$

$$\sin(2\theta) = 2 \sin \theta \cos \theta = 2 \left(\frac{7}{25}\right) \left(-\frac{24}{25}\right) = -\frac{336}{625}$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = \left(-\frac{24}{25}\right)^2 - \left(\frac{7}{25}\right)^2 = \frac{527}{625}$$

$$\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{\left(-\frac{336}{625}\right)}{\left(\frac{527}{625}\right)} = -\frac{336}{527}$$

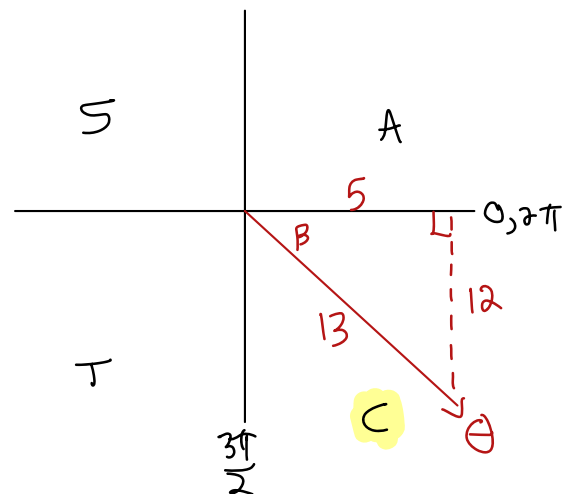


6) Determine the values of $\sin 2\theta$, $\cos 2\theta$, and $\tan 2\theta$, given $\sin \theta = -\frac{12}{13}$ and $\frac{3\pi}{2} \leq \theta \leq 2\pi$

$$\sin(2\theta) = 2 \sin \theta \cos \theta = 2 \left(-\frac{12}{13}\right) \left(\frac{5}{13}\right) = -\frac{120}{169}$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = \left(\frac{5}{13}\right)^2 - \left(-\frac{12}{13}\right)^2 = -\frac{119}{169}$$

$$\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{\left(-\frac{120}{169}\right)}{\left(-\frac{119}{169}\right)} = \frac{120}{119}$$

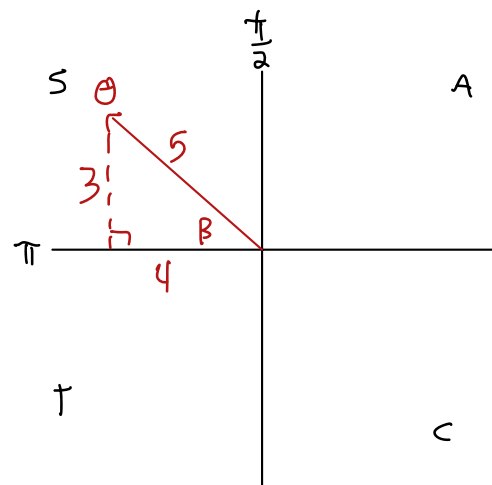


7) Determine the values of $\sin 2\theta$, $\cos 2\theta$, and $\tan 2\theta$, given $\cos \theta = -\frac{4}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$

$$\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(-\frac{4}{5}\right) = -\frac{24}{25}$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = \left(-\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = \frac{7}{25}$$

$$\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{\left(-\frac{24}{25}\right)}{\left(\frac{7}{25}\right)} = -\frac{24}{7}$$



8) Determine the value of a in the equation $2\tan x - \tan(2x) + 2a = 1 - \tan(2x)\tan^2 x$

$$2\tan x + 2a - 1 = \tan(2x) - \tan(2x)\tan^2 x$$

$$2\tan x + 2a - 1 = \tan(2x)(1 - \tan^2 x)$$

$$2\tan x + 2a - 1 = \frac{2\tan x}{1 - \tan^2 x}(1 - \tan^2 x)$$

$$2\tan x + 2a - 1 = 2\tan x$$

$$2a - 1 = 0$$

$$a = \frac{1}{2}$$

Answer Key

1)a) $\sin(10x)$ b) $\cos(2\theta)$ c) $\cos(6x)$ d) $\tan(8x)$ e) $2\sin(2\theta)$ f) $\cos\theta$

2)a) $\sin 90^\circ$; 1 b) $\cos 60^\circ$; $\frac{1}{2}$ c) $\sin \frac{\pi}{6}$; $\frac{1}{2}$ d) $\cos \frac{\pi}{6}$; $\frac{\sqrt{3}}{2}$ e) $\cos \frac{3\pi}{4}$; $-\frac{1}{\sqrt{2}}$ f) $\sin 120^\circ$; $\frac{\sqrt{3}}{2}$

3)a) $2\sin(2\theta)\cos(2\theta)$ b) $2\cos^2(1.5x) - 1$ c) $\frac{2\tan(0.5x)}{1 - \tan^2(0.5x)}$ d) $\cos^2(3\theta) - \sin^2(3\theta)$ e) $2\sin(0.5x)\cos(0.5x)$ f) $\frac{2\tan(2.5\theta)}{1 - \tan^2(2.5\theta)}$

4) $\sin(2\theta) = \frac{24}{25}$, $\cos(2\theta) = -\frac{7}{25}$, $\tan(2\theta) = -\frac{24}{7}$

5) $\sin(2\theta) = -\frac{336}{625}$, $\cos(2\theta) = \frac{527}{625}$, $\tan(2\theta) = -\frac{336}{527}$

6) $\sin(2\theta) = -\frac{120}{169}$, $\cos(2\theta) = -\frac{119}{169}$, $\tan(2\theta) = \frac{120}{119}$

7) $\sin(2\theta) = -\frac{24}{25}$, $\cos(2\theta) = \frac{7}{25}$, $\tan(2\theta) = -\frac{24}{7}$

8) $a = \frac{1}{2}$